

Angelo Theodorou

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Work Experience

April 2024 - To Date, Self Employed (Granada)

- Working full-time on [nCine](#), a cross-platform 2D open source game engine
- Presentations: [14 Years of Developing nCine](#), [The nCine Job System](#)

January 2021 - March 2024, The Multiplayer Group (Remote from Granada)

Role: Senior Rendering Engineer

Description: Worked on Bethesda's Starfield as a Senior Rendering Engineer.

Technologies: C++, Direct3D 12, Vulkan, HLSL, GLSL

Tools: Visual Studio, Perforce, RenderDoc, PIX, Nsight Graphics

Website: <https://themultiplayergroup.com/>

March 2019 - December 2020, Self Employed (Granada)

- Worked full-time on [nCine](#), a cross-platform 2D open source game engine
- Developed [SpookyGhost](#), a procedural animation tool for sprites made with nCine and ImGui

March 2017 - February 2019, Frostbite (Stockholm)

Role: Rendering Engineer

Description: Worked in the Image Quality engine rendering team. Some of my tasks have been:

- Help with the DirectX 12 RTX integration in the editor
- Add a way to capture a sequence of screenshots to test temporal effects stabilization
- Transition textured and frustum lights from deferred/stencil to tiled lighting
- Add support for newer versions of Visual Studio in the custom Enlighten build system
- Initial analysis of the integration effort to merge back Bioware's Anthem Enlighten changes

Technologies: C++, Direct3D, HLSL

Tools: Visual Studio, Perforce, RenderDoc

June 2015 - February 2017, NaturalMotion Ltd. (Oxford)

Role: Android Technology Programmer

Description: Worked in the Core Game Tech team on the in-house Echo engine used for games like *Dawn of Titans* and *Clumsy Ninja*. Some of my tasks have been:

- Initial porting of the engine from OpenGL to Vulkan
- Rewrite the API for atomic operations on all supported platforms
- Refactor the ARM NEON intrinsics code for vector and matrix operations with unit tests
- Help with the transition from Android Lollipop to Marshmallow

I also worked with the CSR2 team to migrate their building process from Ant to Gradle.

Technologies: C++, Java, OpenGL ES, GLSL, Vulkan, Android API

Tools: Visual Studio, Subversion, git

December 2012 - June 2015, ARM Ltd. (Cambridge)

Role: Senior Software Engineer

Description: Worked in the Middleware Graphics team in Mali Ecosystem for the Media Processing Group (MPG) to help game companies achieve the best on Mali GPUs.

My role has been a combination of graphics and game programming, technical support to game middleware companies, presentations at events worldwide and real-time applications profiling and optimization:

- Helped porting Unreal Engine 4 to ARM 64-bit
- Wrote the codebase for a Unity demo revealed at Unite 2014
- Presented sponsored sessions at GDC, Unite, and Gamelab Barcelona
- Assisted companies like Frostbite, Epic Games, Unity, and Gameloft with the ARM and Mali tools

Technologies: C++, C#, Python, OpenGL ES, GLSL

Tools: Visual Studio, Qt Creator, CMake, git, Unity

Jun 2010 - November 2012, SpinVector (Benevento)

Role: Programmer

Description: Worked as a game programmer and a multi-platform GUI programmer on three released titles: *From Cheese*, *AXL: Full Boost*, and *Bang!: The Official Videogame*.

Technologies: C++, Qt4, Objective C, UIKit, Java, Android API

Tools: Visual Studio, Xcode, Qt Creator, qmake, Eclipse, Subversion, Unity

Education

- Bachelor's Degree in Computer Science, University of Naples "Federico II", awarded 110/110 cum laude in 2009